

Variance Reduction Note 2: The Convergence of SPIDER

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Abstract

The stochastic path-integrated differential estimator (SPIDER) tracks a gradient by accumulating small, shared-sample gradient differences and periodically refreshing the estimate. This second expository note on recursive variance reduction explains SPIDER-SFO and gives a self-contained proof of its expected first-order stationarity guarantee. The argument combines an epochwise mean-square error bound with descent under a clipped, normalized update. It yields $O(\varepsilon^{-3})$ stochastic gradient evaluations in the online setting and $O(n + \sqrt{n}\varepsilon^{-2})$ component gradient evaluations for a finite sum, with the remaining problem parameters fixed. All refresh costs are included. The presentation is pedagogical and makes no claim of a new algorithm or convergence rate.

1 From recursive estimates to periodic refreshes

Consider a smooth, possibly nonconvex objective

$$\min_{x \in \mathbb{R}^d} F(x) := \mathbb{E}_{\xi}[f(x; \xi)], \quad g(x; \xi) := \nabla f(x; \xi). \quad (1)$$

Variance reduction exploits relationships among stochastic gradients rather than estimating each gradient independently. Classical examples include SVRG and SAGA [6, 8]; SARAH introduced a recursive gradient estimator [23]. SPIDER, developed by Fang et al. [26], combines recursive differences with periodic refreshes and controlled iterate movement. Its name emphasizes a discrete path identity: for any epoch starting at s ,

$$\nabla F(x_t) = \nabla F(x_s) + \sum_{k=s+1}^t [\nabla F(x_k) - \nabla F(x_{k-1})].$$

SPIDER estimates the initial term accurately and each subsequent difference cheaply. Using the same sample at both points makes the difference noise small when the points are close.

We study the expectation version of SPIDER-SFO [26], with its free parameter $n_0 = 1$ and integer batch sizes. The proof uses explicit, convenient constants. This complements the momentum-based mechanism of STORM [33]: SPIDER controls accumulated error through refreshes, whereas STORM damps old error within its recursion. For broader context, see [2].

2 The oracle and the SPIDER-SFO update

Assumption 1. *The differentiable function F is bounded below by $F_{\star} > -\infty$. For some $L > 0$ and every $x, y \in \mathbb{R}^d$,*

$$\mathbb{E}[g(x; \xi)] = \nabla F(x), \quad \mathbb{E} \|g(x; \xi) - g(y; \xi)\|^2 \leq L^2 \|x - y\|^2. \quad (2)$$

In the online setting, also assume that, for some $\sigma \geq 0$,

$$\mathbb{E} \|g(x; \xi) - \nabla F(x)\|^2 \leq \sigma^2. \quad (3)$$

All new samples are independent of the past and of one another. A sample can be reused to evaluate its gradient at two specified points. Norms are Euclidean.

The mean-square smoothness condition (2) implies that F is L -smooth by Jensen's inequality. For a finite sum $F(x) = n^{-1} \sum_{i=1}^n f_i(x)$, take ξ uniform on $\{1, \dots, n\}$ and $g(x; i) = \nabla f_i(x)$. The same mean-square condition is imposed, but (3) is unnecessary because refreshes will use the exact full gradient.

Fix x_0 , an internal tolerance $\delta > 0$, a horizon $T \geq 1$, an epoch length $q \geq 1$, and batch sizes $B, b \geq 1$. For $t = 0, \dots, T-1$, first construct v_t as follows:

$$v_t = \begin{cases} \frac{1}{B} \sum_{i=1}^B g(x_t; \zeta_{t,i}), & t \equiv 0 \pmod{q}, \\ v_{t-1} + \frac{1}{b} \sum_{i=1}^b [g(x_t; \xi_{t,i}) - g(x_{t-1}; \xi_{t,i})], & t \not\equiv 0 \pmod{q}. \end{cases} \quad (4)$$

In the finite-sum setting, replace the first line by $v_t = \nabla F(x_t)$, at a cost of n component gradients; sample inner indices independently with replacement. Both evaluations in each inner difference use the same sample. Then update

$$\alpha_t = \frac{\delta}{L \max\{\|v_t\|, 2\delta\}}, \quad x_{t+1} = x_t - \alpha_t v_t. \quad (5)$$

Return x_R , where R is independent of the run and uniform on $\{0, \dots, T-1\}$. This is the clipped normalized step in SPIDER-SFO:

$$0 < \alpha_t \leq \frac{1}{2L}, \quad \|x_{t+1} - x_t\| \leq \frac{\delta}{L}. \quad (6)$$

The maximum in (5) also defines the update when $v_t = 0$. The estimator is generally conditionally biased between refreshes; the analysis below tracks its error explicitly.

3 A short convergence proof

Let $e_t = v_t - \nabla F(x_t)$ and $Q_t = \mathbb{E} \|e_t\|^2$. Write $s = q \lfloor t/q \rfloor$ for the most recent refresh index. At every refresh, $Q_s \leq \nu^2$, where $\nu^2 = \sigma^2/B$ online and $\nu^2 = 0$ for an exact finite-sum refresh.

Lemma 1 (Error accumulation within an epoch). *Under Assumption 1, the recursion satisfies*

$$Q_t \leq \nu^2 + \frac{L^2}{b} \sum_{k=s+1}^t \mathbb{E} \|x_k - x_{k-1}\|^2 \leq \nu^2 + \frac{q\delta^2}{b}. \quad (7)$$

The sum is zero when $t = s$.

Proof. For a nonrefresh step k , let $\mathcal{H}_k$ contain all randomness before drawing its new batch, so x_k, x_{k-1}, v_{k-1} are measurable. Define the centered increment

$$Z_k = \frac{1}{b} \sum_{i=1}^b [g(x_k; \xi_{k,i}) - g(x_{k-1}; \xi_{k,i})] - [\nabla F(x_k) - \nabla F(x_{k-1})].$$

Then $e_k = e_{k-1} + Z_k$ and $\mathbb{E}[Z_k \mid \mathcal{H}_k] = 0$. Independence within the batch and the variance identity give

$$\mathbb{E}[\|Z_k\|^2 \mid \mathcal{H}_k] \leq \frac{1}{b} \mathbb{E}[\|g(x_k; \xi) - g(x_{k-1}; \xi)\|^2 \mid \mathcal{H}_k] \leq \frac{L^2}{b} \|x_k - x_{k-1}\|^2.$$

The cross term with e_{k-1} vanishes conditionally, hence $Q_k \leq Q_{k-1} + (L^2/b) \mathbb{E} \|x_k - x_{k-1}\|^2$. Summing from $s+1$ to t and using (6) proves (7). At a sampled refresh, the independent, centered errors average to variance at most σ^2/B . $\square$

Lemma 2 (Descent with a controlled estimator). *If $Q_t \leq \delta^2$, then*

$$\mathbb{E}F(x_{t+1}) \leq \mathbb{E}F(x_t) - \frac{\delta}{4L} \mathbb{E} \|v_t\| + \frac{\delta^2}{2L}. \quad (8)$$

Proof. Since $\nabla F(x_t) = v_t - e_t$, smoothness and $\langle e_t, v_t \rangle \leq (\|e_t\|^2 + \|v_t\|^2)/2$ yield

$$\begin{aligned} F(x_{t+1}) &\leq F(x_t) - \alpha_t \|v_t\|^2 + \alpha_t \langle e_t, v_t \rangle + \frac{L\alpha_t^2}{2} \|v_t\|^2 \\ &\leq F(x_t) - \frac{\alpha_t}{4} \|v_t\|^2 + \frac{1}{4L} \|e_t\|^2. \end{aligned}$$

Here (6) bounds the coefficients pointwise; α_t need not be independent of e_t . For every $r \geq 0$, $r^2/\max\{r, 2\delta\} \geq r - \delta$. For $r \leq 2\delta$, this follows from $r^2 - 2\delta r + 2\delta^2 = (r - \delta)^2 + \delta^2 \geq 0$; for $r \geq 2\delta$, it is immediate. Thus $\alpha_t \|v_t\|^2 \geq (\delta/L)(\|v_t\| - \delta)$. Taking expectations and using $Q_t \leq \delta^2$ proves the result. $\square$

Theorem 1 (Expected stationarity). *Let $\Delta = F(x_0) - F_\star$. If the refresh and batch choices ensure $\nu^2 + q\delta^2/b \leq \delta^2$, then SPIDER-SFO satisfies*

$$\mathbb{E} \|\nabla F(x_R)\| \leq \frac{4L\Delta}{\delta T} + 3\delta. \quad (9)$$

Consequently, for any target $\varepsilon > 0$, choose

$$\delta = \frac{\varepsilon}{4}, \quad T = \max \left\{ 1, \left\lceil \frac{4L\Delta}{\delta^2} \right\rceil \right\}. \quad (10)$$

Then $\mathbb{E} \|\nabla F(x_R)\| \leq \varepsilon$.

Proof. Lemma 1 gives $Q_t \leq \delta^2$ at every iteration. Summing Lemma 2 and using $F(x_T) \geq F_\star$ gives

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|v_t\| \leq \frac{4L\Delta}{\delta T} + 2\delta.$$

By the triangle and Cauchy–Schwarz inequalities, $\mathbb{E} \|\nabla F(x_t)\| \leq \mathbb{E} \|v_t\| + \sqrt{Q_t}$. Averaging over the independent uniform index R proves (9); (10) makes its right-hand side at most $4\delta = \varepsilon$. $\square$

Online oracle complexity. Choose

$$B = \max\{1, \lceil 2\sigma^2/\delta^2 \rceil\}, \quad q = \max\{1, \lceil \sigma/\delta \rceil\}, \quad b = 2q. \quad (11)$$

Then $\sigma^2/B \leq \delta^2/2$ and $q\delta^2/b = \delta^2/2$, so Theorem 1 applies. There are $m = \lceil T/q \rceil$ refreshes, including the initial one. Each inner step uses $2b$ gradient evaluations. Therefore the total is

$$N_{\text{on}} = mB + 2b(T - m) \leq B + (B/q + 2b)T = O\left(1 + \frac{\sigma^2}{\varepsilon^2} + \left(1 + \frac{\sigma}{\varepsilon}\right) \frac{L\Delta}{\varepsilon^2}\right). \quad (12)$$

This is $O(\varepsilon^{-3})$ for fixed L, Δ, σ as $\varepsilon \downarrow 0$. The ceilings also cover $\sigma = 0$, when $q = B = 1$ and each step uses an exact gradient almost surely.

Finite-sum oracle complexity. Use exact refreshes and take $q = b = \lceil \sqrt{n} \rceil$. Now $\nu = 0$ and $q\delta^2/b = \delta^2$, so the same theorem applies without a bounded-variance assumption. The component-gradient count is

$$N_{\text{fin}} = mn + 2b(T - m) \leq n + (n/q + 2b)T = O\left(n + \frac{\sqrt{n} L\Delta}{\varepsilon^2}\right). \quad (13)$$

Indeed, $n/q \leq \sqrt{n}$ and $b = q \leq 2\sqrt{n}$; the extra $O(\sqrt{n})$ from rounding T is absorbed by n .

Interpretation and scope. The proof separates three roles: normalization bounds movement, shared samples control the variance of each difference, and refreshes limit how many such errors accumulate. Balancing the amortized refresh cost with the inner batch cost gives the stated oracle bounds. The schedule uses L , the target accuracy, and online a variance bound; choosing T also uses Δ or an upper bound on it. The result concerns the expected gradient norm of a random iterate, not a squared-norm, last-iterate, high-probability, or second-order guarantee. Constant-step alternatives such as SpiderBoost [28] require a different descent argument. STORM and its adaptive variants [33, 35, 37] offer a complementary way to control recursive estimation error.

4 Recommended reading

Variance-reduction methods. Basic guarantees [1–3]; gradient memories and control variates [4–10]; sampling and acceleration [11–19]; nonconvex and recursive estimators [20–32]; momentum and adaptive schedules [33–43]; second-order extensions [44–47]; lower bounds [48].

Compositional optimization. Two-level approximation [49–52]; finite-sum and recursive estimators [53–56]; multilevel tracking [57–60]; coupled and partially observed blocks [61–65]; projection-free methods [66–70]; statistical-model applications and federated extensions [71–74].

Sign-based optimization. Quantization, signSGD, majority voting, and error feedback [75–79]; variance-reduced sign updates and Lion [80–82]; compressed gradient tracking [83–87]. Related distributed control-variate methods include SCAFFOLD, FedPAGE, decentralized hybrid estimators, and ProxSkip [88–91].

Minimax optimization. Finite-sum saddle-point methods, matrix games, and variance-reduced extragradient [92–95]; recursive, hybrid, adaptive, and decentralized nonconvex minimax methods [96–99]; a variational-inequality treatment [100].

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